RAK12025 WisBlock Gyroscope Sensor Module Datasheet

Overview

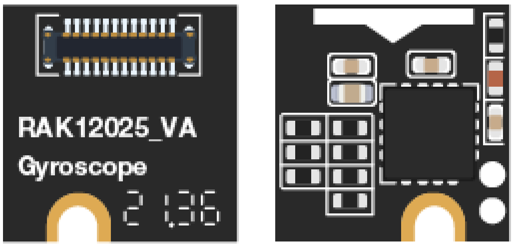

Figure 1: RAK12025 WisBlock Gyroscope Sensor Module

Description

RAK12025 is a gyroscope module, part of the RAKWireless WisBlock Sensor series. The module is based on I3G4250D from STMicroelectronics. The I3G4250D is a low-power 3-axis angular rate sensor able to provide unprecedented stability at a zero-rate level and sensitivity over temperature and time. It includes a sensing element and a digital interface capable of providing the measured angular rate. With I3G4250D, RAK12025 can measure rotation speed and report data through a standard I2C digital interface.

Features

- **Gyroscope module**
- **Selectable full scale:** (245/500/2000 dps)
- **I2C interface**
- **16-bit rate value data output**
- **8-bit temperature data output**
- **Module size**: 10 × 10 mm
- **Power Supply Voltage**: 3.3 V
- **Current Consumption**: 5 uA - 6.1 mA
- **Chipset**: STMicroelectronics I3G4250D

Specifications

Overview

Mounting

Figure 1 shows the mounting mechanism of the RAK12025 module on a WisBlock Base board. The RAK12025 module can be mounted on the slots: **C, D, E, & F**.

> **NOTE**
>
> RAK12025 has two digital output lines, so you need two GPIOs from WisBlock Core. It means RAK12025 should be used on a sensor slot with two available GPIOs. However, WB_IO2 is used to control 3V3_S. Hence, RAK12025 is used only on slots without WB_IO2 like sensor slots C and D on WisBlock Base board.

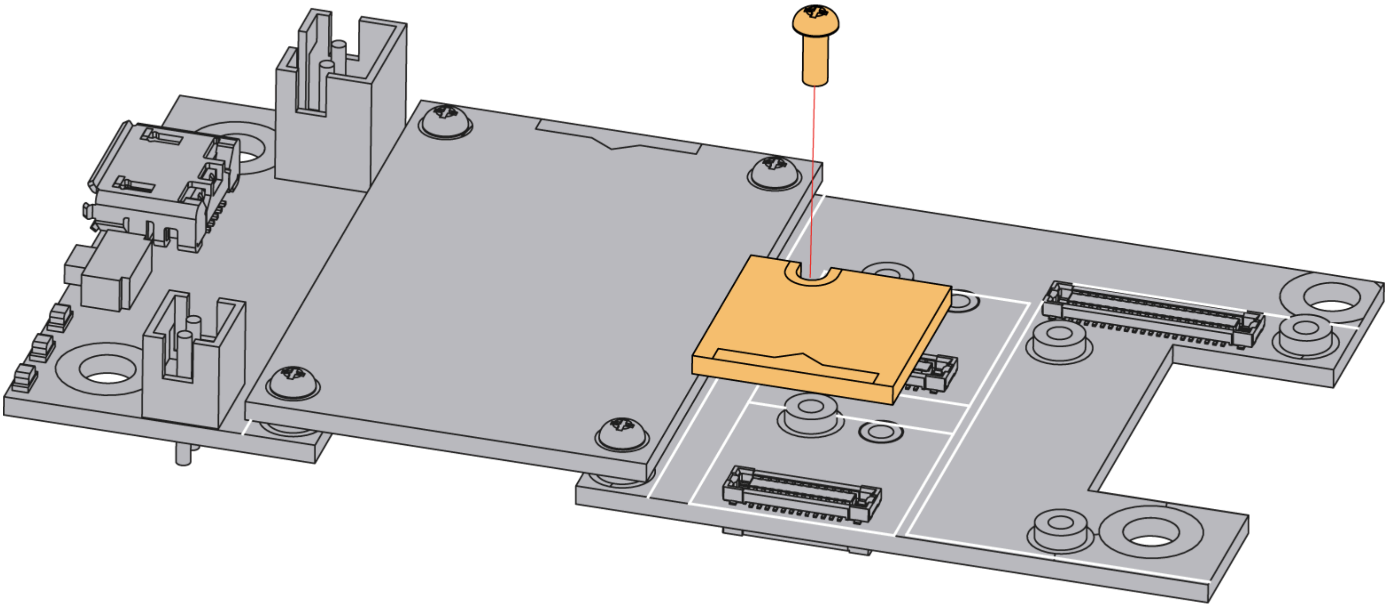

Figure 2: Mounting of RAK12025

Hardware

The hardware specification is categorized into six (6) parts. It shows the chipset of the module and discusses the pinouts, sensors, and the corresponding functions and diagrams. It also covers the electrical and mechanical parameters that include the tabular data of the functionalities and standard values of the RAK12025 WisBlock Gyroscope Sensor Module.

Chipset

Vendor	Part number
STMicroelectronics	I3G4250D

Pin Definition

The RAK12025 WisBlock Gyroscope Sensor Module comprises a standard WisBlock connector. The WisBlock connector allows the RAK12025 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition is shown in **Figure 3**.

> **NOTE**
>
> **I2C** related pin, **INT2**, **INT1**, **3V3_S**, and **GND** are connected to Wisblock connector.

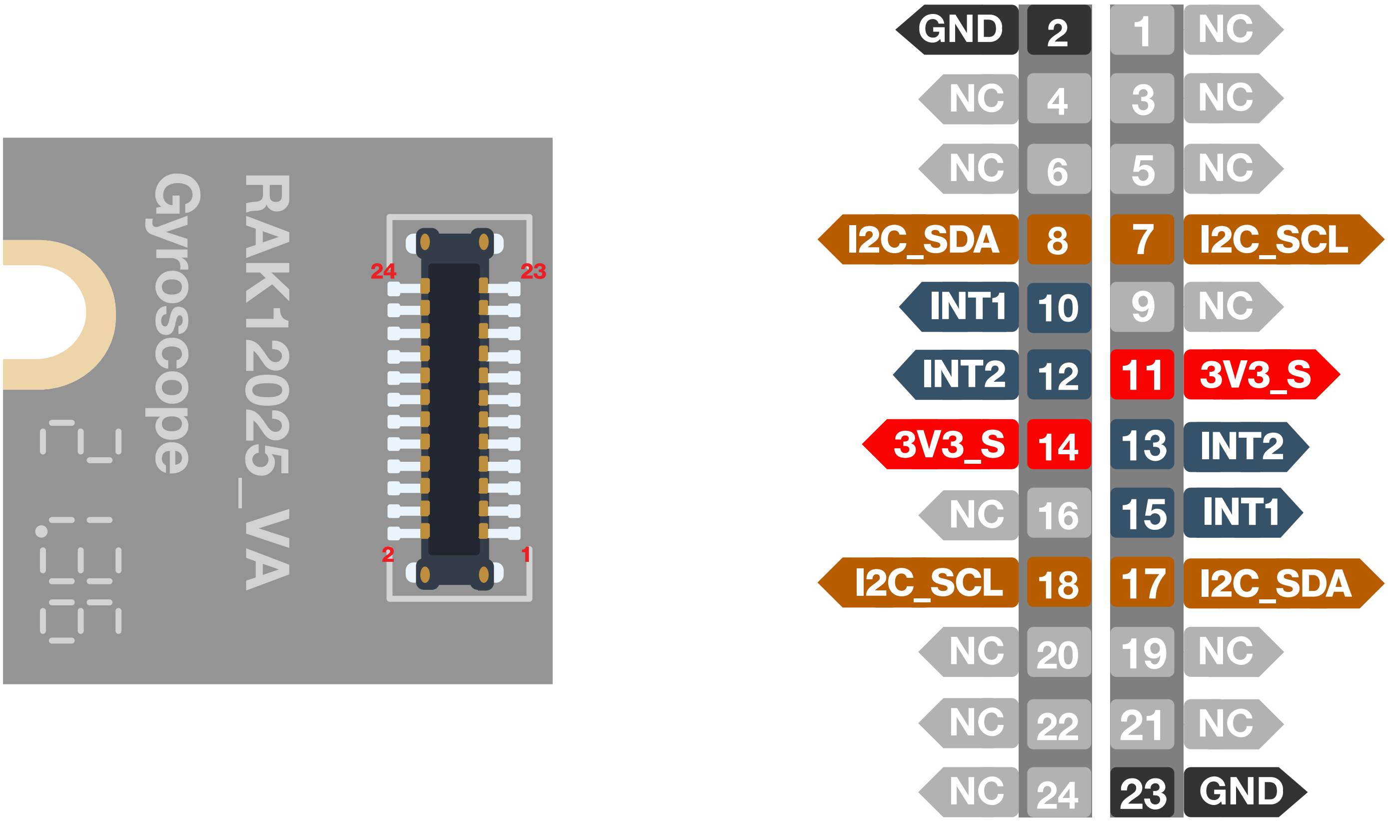

Figure 3: RAK12025 pinouts

If a 24-pin WisBlock Sensor connector is used, the IO used for the output pulse depends on what slot the module is plugged in. The following table shows the default IO used for different slots:

The table below shows the default IOs used for different slots using INT1:

SLOT C	SLOT D	SLOT E	SLOT F
WB_IO4	WB_IO6	WB_IO3	WB_IO5

When using INT2, the default IOs used for different slots is shown in the table below:

SLOT C	SLOT D	SLOT E	SLOT F
WB_IO3	WB_IO5	WB_IO4	WB_IO6

Sensors

Gyroscope Sensor

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Unit
FS	Measurement Range	User-selectable		±245		dps
		User-selectable		±500		dps
		User-selectable		±2000		dps
So	Sensitivity	FS bit = 245 dps	7.4	8.75	10.1	mdps/digit
		FS bit = 500 dps	14.8	17.50	19.8	mdps/digit

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Unit
		FS bit = 2000 dps	59.2	70	79.3	mdps/digit
SoDr	Sensitivity change vs. Temperature	From -40 °C to +85 °C		±2		%
DVoff	Digital zero-rate level	FS = 245 dps	-25	±10	+25	dps
		FS = 500 dps	-37.5	±15	+37.5	dps
		FS = 2000 dps	-187.5	±75	+187.5	dps
OffDr	Zero-rate level change vs. Temperature	FS = 245 dps		±0.03		dps/°C
NL	Non-linearity	Best fit straight line	-5	0.2	+5	dps/°C
DST	Self-test output change	FS = 245 dps		130		dps
		FS = 500 dps		200		dps

Symbol	Parameter	Test Condition	Min.	Typ.	Max.	Unit
		FS = 2000 dps		530		dps
Rn	Rate noise density	BW = 50 Hz		0.03		dps/sqrt (Hz)
ODR	Digital output data rate			105/208/420/840		Hz
Top	Operating temperature range		-40		+85	°C

Electrical Characteristics

- @ Vdd = 3.0 V, T = +25 °C, unless otherwise noted.

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
Vdd	Supply voltage	Input voltage must within this range	2.4	3.3	3.6	V
Idd	Supply current	Normal	-	6.1	-	mA
IddSL	Supply current in sleep mode	Sleep mode	-	1.5	-	uA
IddPdn	Supply current in power-down mode	Power-down mode		5		uA

Symbol	Description	Condition	Min.	Typ.	Max.	Unit
Top	Operating temperature range		-40		+85	°C

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and the mechanic drawing of the RAK12025 module.

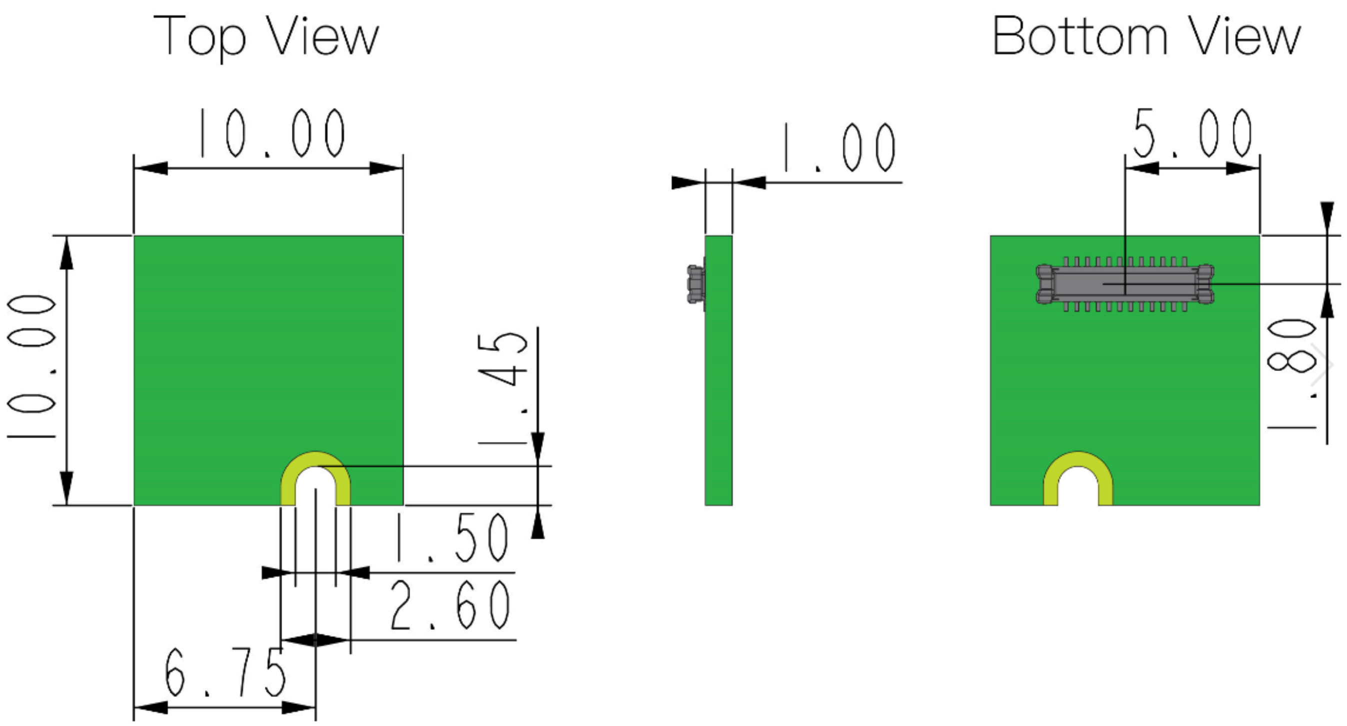

Figure 4: RAK12025 mechanical dimensions

WisConnector PCB Layout

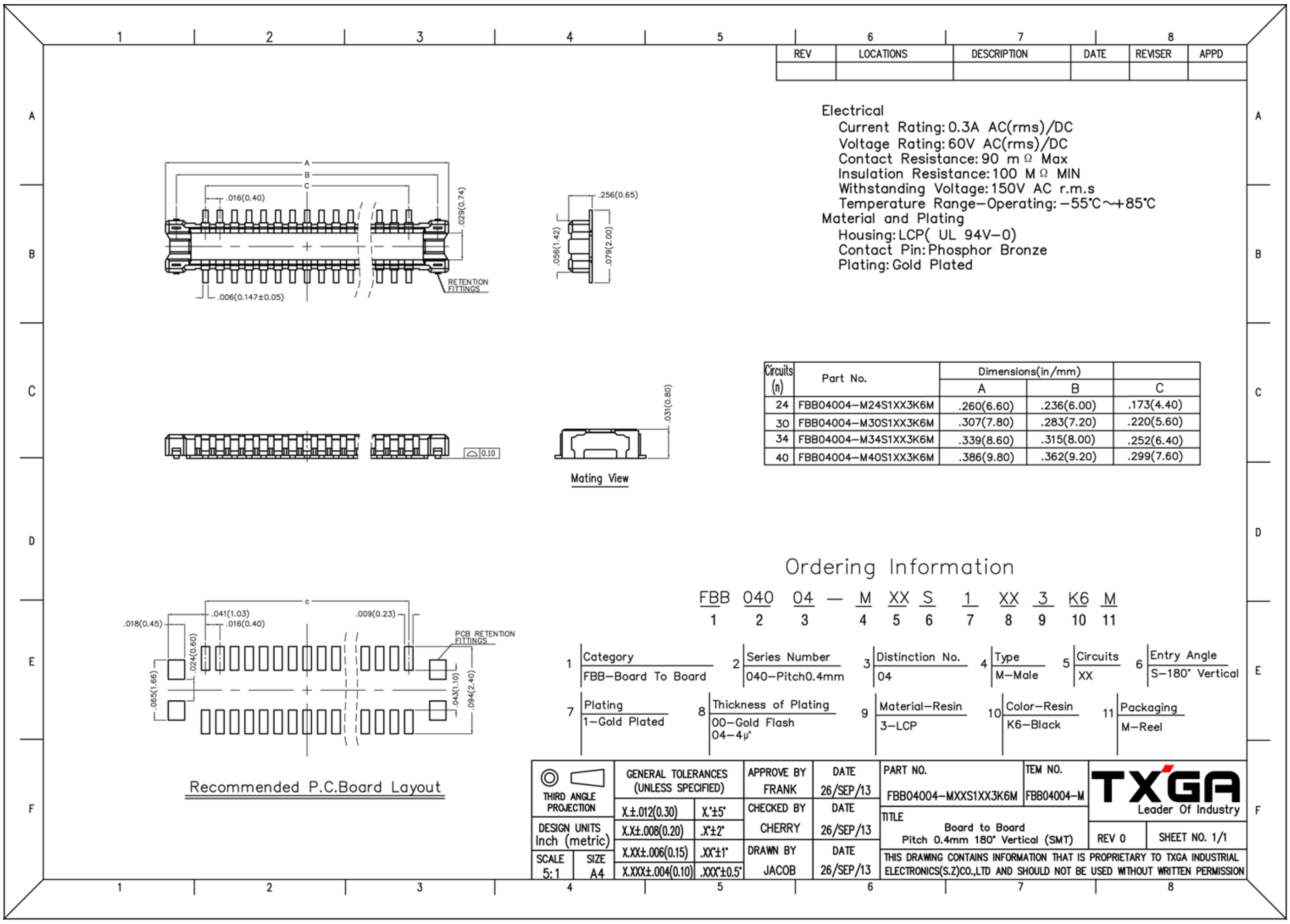

Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

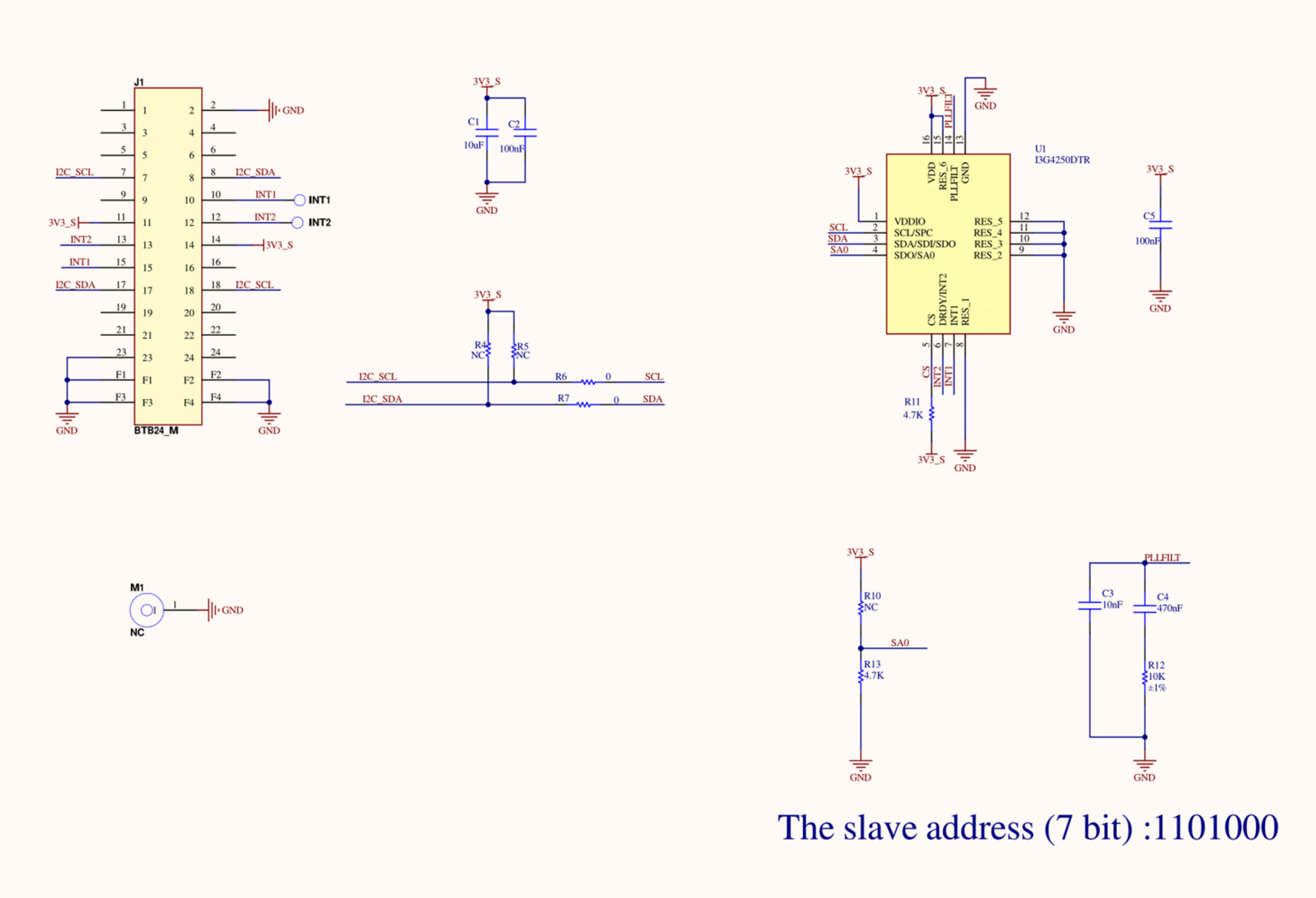

Figure 6: RAK12025 schematic diagram